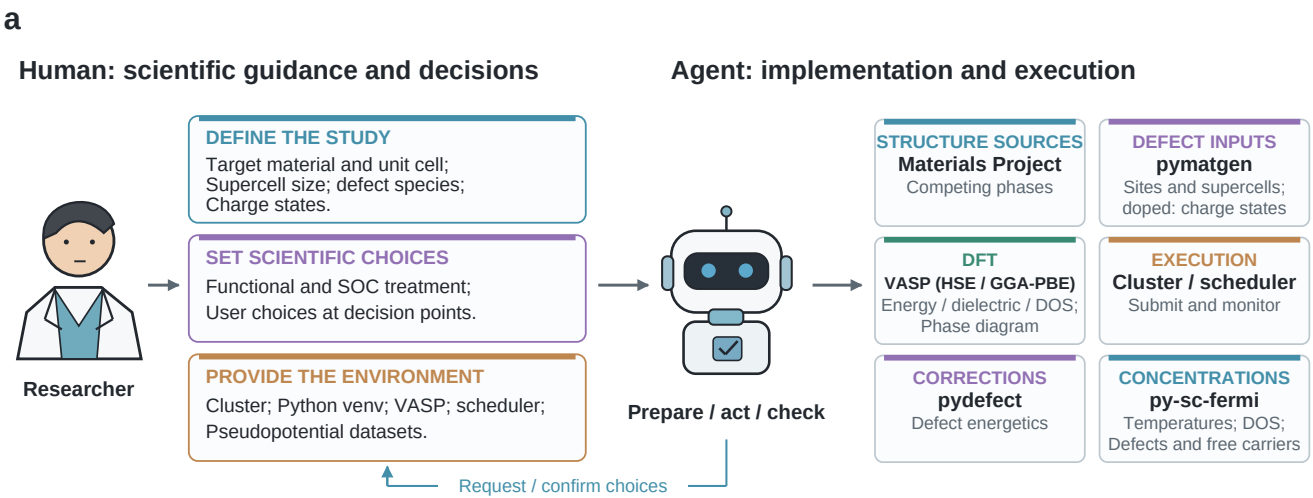

b Full defect calculation and analysis protocol

HOST REFERENCES	DEFECT PREPARATION	DEFECT CALCULATIONS	RESULTS AND BACKUP
1 **Unit-cell structure and electronic structure** Relaxation; band gap; DOS	4 **Supercell and defects** Symmetry-inequivalent sites; charge states; site multiplicities	7 **Defect relaxation** Fixed lattice; spin polarization; symmetry off	10 **Formation-energy plots** At every chemical-potential vertex
2 **Dielectric constant** Electronic and ionic contributions	5 **Validate and execute** Check inputs; submit and monitor	8 **Static calculations** Defects and defect-free supercell	11 **Defect thermodynamics** Temperature; charge neutrality; Defect and carrier concentrations
3 **Phase stability** Formation enthalpies; Elemental chemical potentials	6 **Interstitial screening** GGA-PBE; select distinct minima	9 **Defect postprocessing** Finite-size corrections; formation energies at the VBM	12 **Data backup** Preserve hierarchy; verify checksums

◇ **Shared control at every calculation stage**
Validate inputs, check convergence, and resume unfinished relaxations in continue_N directories.

○ User decision / confirmation ◇ Validation / recovery checkpoint